The Single-Site T-Matrix of a Legendre Cell in Closed Form

Elastic Waves, Every Power of the Frequency on Six Constants

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Draft, 7 October 2026

Abstract

A volume-integral method for elastic waves divides a heterogeneous medium into cells, each with a single-site T-matrix that maps the field incident on it to the sources it radiates when it is alone in the background medium. The governing equation is written in displacement; how the strain inside a cell is represented is a modelling decision. The classical approximation, of Eshelby's equivalent-inclusion method and of the coupled-dipole method for elastic waves, takes it as uniform in the cell. We expand the displacement and the strain independently in Legendre polynomials over the cube and test against the same polynomials: a Galerkin cell. We choose it because its order is cheap: it is no more accurate than a point-matching cell of the same size and degree, but raising its degree raises its order at small cost in unknowns, so that for a given accuracy it needs fewer unknowns than point matching, which imposes the equation at the cell's centre and treats the receiving cell as a point. The price is that its interactions are double integrals over both cells, singular where cells touch. For the degree-one Legendre cell, whose expansions hold the constant and the three linear polynomials (36 unknowns), with a uniform contrast we give the single-site T-matrix in closed form at every power of the wavenumber. Its self-interaction is a power series, $\mathbf{K} = \sum_n (k_S h)^n \mathbf{K}_n$ with $k_S = \omega/\beta$ the shear wavenumber of the background (ω the angular frequency, β the shear-wave speed) and h the half-width of the cubic cell, whose coefficients $\mathbf{K}_n$ are moments of powers of the distance over the cell and depend on neither the frequency, the cell size nor the medium. Every coefficient is exact: an even power of $k_S h$ is a combination with rational coefficients of six constants, 1, π, $\sqrt{2}$, $\sqrt{3}$, $\ln(1+\sqrt{2})$ and $\ln(2+\sqrt{3})$, the logarithms coming from the diagonals of the cube's faces and of the cube itself, and an odd power, which carries the radiation, is rational. The first of these is the radiation damping of a point force, recovered exactly. On the irreducible representations of the cube group the 36 unknowns of the cell separate into blocks of at most four, each inverted in closed form: the T-matrix is an explicit rational function of the contrast and of $k_S h$, whose poles are written down. In the static limit the cell separates into independent scalar channels, each of Eshelby's form $MD/(1-ND)$ with one closed-form constant N per channel; the dilatational channel is Eshelby's sphere exactly, and the two shear channels average to the sphere's. Through $(k_S h)^6$ the closed form reproduces the T-matrix computed by quadrature to 3.5×10^{-9} at $k_S h = 0.2$, falling as $(k_S h)^7$ as the next term predicts, and each of its 223 moments to 1.7×10^{-15}. Because the coefficients depend on neither the frequency, the cell size nor the medium, a frequency sweep integrates nothing

at any frequency: the T-matrix costs 0.43 ms per frequency against 1.2 s by quadrature, after a one-off derivation of 30 s.

1 Introduction

A volume-integral method for elastic scattering replaces a heterogeneous medium by cells, here cubes, and couples them through the Green's tensor of the background. Each cell is characterised by its single-site T-matrix: the map from the field incident on it to the sources it radiates when it is alone in the background medium. The coupled system is then the multiple-scattering problem of the cells [1–4], and everything that a cell contributes to the solution, beyond its coupling to the others, is in its T-matrix.

Why an integral equation. The volume-integral equation (the Lippmann–Schwinger equation of elastodynamics) does not solve for the wave in the heterogeneous medium directly. It treats the heterogeneity as a set of secondary sources radiating into a background whose response is known: a density contrast acts as a body force proportional to the local displacement, a stiffness contrast as a stress source proportional to the local strain, and the scattered field is the background's response to these sources [5, 6]. Each step of a solution then has a physical reading. The first term in the contrast is single scattering, the Born approximation; the following terms count the interactions between parts of the body; and a cell's T-matrix is its equivalent source, a force and a moment tensor of the kind used to describe earthquakes [7], so the cell's scattering can be read channel by channel, as §5 does. The radiation condition is built into the Green's tensor, so only the heterogeneous region is discretised, with no absorbing boundary to tune; and the transition operator that results is the natural object for inversion, where the contrast is the unknown [8, 9].

What it costs. The same formulation has difficulties of its own, and they shape this series of papers. The Green's tensor is singular where source and receiver meet, and its second derivatives, which carry the strain, are strongly singular, as r^{-3} in three dimensions, integrable only as a principal value with a delta term: the interaction of a cell with itself and with its neighbours is a singular integral that ordinary quadrature cannot evaluate [10], and it is the largest entry of the system. Every cell interacts with every other, so the system is dense, and its cost grows as the square of the number of cells unless a fast method is used: fast Fourier transforms on regular grids [11, 12], their corrected and adaptive forms [13, 14], or fast multipoles [15]. The discretisation of a strongly singular operator is not automatically stable [16], and at high contrast the multiple-scattering series diverges, so the equation must be solved rather than expanded. The background must have a Green's tensor in usable form, which in practice means a homogeneous or layered medium. Cubes represent curved boundaries by staircases. And the method works at one frequency at a time, so a broadband calculation repeats the singular integrals at every frequency. This paper removes the first and the last of these for the single cell: its self-interaction is obtained in closed form, at every power of the frequency, once. Nestor [17] does the same for the coupling between cells, and Nestor [18] treats the error of the discretisation.

How the field inside a cell is represented decides what that T-matrix is. The governing

equation is written in displacement, and the strain enters it only through the stiffness contrast; how the strain is represented within a cell is therefore a modelling decision. The classical approximation, of the equivalent-inclusion method [19, 20] and of the coupled-dipole method for elastic waves [21], takes it as uniform. That approximation is relaxed here by expanding the field in each cell in Legendre polynomials and testing the equation against the same polynomials (a Galerkin cell). The strain can then be represented in two ways: as the derivative of each cell's displacement polynomial, or as an unknown in its own right, with its own Legendre expansion over the cell. Which strain a tiling of cells needs is measured on a layer, which has an exact solution (Table 2). Both schemes there are Galerkin, with the displacement a polynomial of degree p in each cell; they differ only in the strain. The error in the reflected wave is followed as the cells are refined, and its order is the power of the cell size h at which it falls. With the strain an independent polynomial of degree p, the error falls as h^{2p+2}: as h^4, h^6 and h^8 for $p = 1$, 2 and 3. With the strain taken as the derivative of each cell's displacement polynomial, it falls only as h^2, h^2 and h^4, so that raising the degree from one to two gains nothing.

This paper takes the second way, the Galerkin cell whose strain is an independent unknown expanded in Legendre polynomials. The reason to prefer a Galerkin cell is not that it is more accurate cell for cell: a point-matching cell of the same degree, which imposes the equation at its centre and treats the receiving cell as a point, has an error of the same order with a different constant [18]. It is that its order is cheap. Raising the degree of its polynomials raises its order at small cost in unknowns, whereas point matching raises its order only by imposing the gradients of the equation at the centre as well: fourth order costs the Galerkin cell 36 unknowns and point matching 60, and on a graded sphere the Galerkin cell of degree one needs 6.6 to 24 times fewer unknowns than the classical uniform point cell for the same accuracy [18]. The price is in the integrals. The interactions of a Galerkin cell are double integrals over both cells, singular where cells touch, which is where numerical integration is weakest; here and in Nestor [17] they are exact, and evaluated to round-off even for cells that touch. And a Galerkin cell holds the projection of the field and of the medium, which is what makes its error a projection error, known before the solve [18].

The single-site T-matrix of the degree-one Legendre cell was given in Nestor [17] in its static limit, entry by entry on five constants. Here it is given at every power of the frequency. The central result is that the self-interaction of the cell, the only part of its T-matrix that is an integral, is

$$\mathbf{K} = \sum_{n\geq 0} (k_S h)^n \, \mathbf{K}_n, \qquad \mathbf{K}_n \text{ exact for every } n, \tag{1}$$

with $k_S = \omega/\beta$ the shear wavenumber of the background and h the half-width of the cell: a power series in $k_S h$ whose coefficients are universal moments of powers of the distance over the cell [17], independent of the frequency, the cell size and the medium. We evaluate every one of them exactly (§3), and find that each even power of $k_S h$ is a combination with rational coefficients of the same six constants and that each odd power, which carries the radiation, is rational (§4). Computed once and stored, the $\mathbf{K}_n$ give the T-matrix at any frequency without an integral (§7). The cube group reduces the 36 unknowns to blocks of at most four (§5, Figure 2), on which the T-matrix is a few closed-form matrices per power, each inverted in closed form; in the static limit

the cell is a set of independent channels of Eshelby's form, one constant each (Table 4). Section 6 gives the checks, each against an evaluation that shares no step with the closed form.

2 The cell and its single-site matrix

The governing equation. With time dependence $\mathrm{e}^{-\mathrm{i}\omega t}$, split the density and the stiffness into the background and the contrast, $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, $\delta c_{bcde} = \Delta\lambda\,\delta_{bc}\delta_{de} + \Delta\mu(\delta_{bd}\delta_{ce} + \delta_{be}\delta_{cd})$. The contrast acts on the background as a body force, $\omega^2\Delta\rho\,u_b + \partial_c m_{bc}$ with $m_{bc} = \delta c_{bcde}\varepsilon_{de}$, and with $\mathbf{G}$ the background's Green's tensor the displacement is

$$u_a(\mathbf{x}) = u^0_a(\mathbf{x}) + \int G_{ab}(\mathbf{x}-\mathbf{y})\,\omega^2\Delta\rho\,u_b(\mathbf{y})\,\mathrm{d}^3y + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y})\,m_{bc}(\mathbf{y})\,\mathrm{d}^3y, \tag{2}$$

the moment radiating through the receiver derivative of $\mathbf{G}$ [18]. The strain enters only through m. Differentiating Eq. (2) gives the strain at $\mathbf{x}$, and the pair $\psi = (\mathbf{u}, \boldsymbol{\varepsilon})$, nine components with the strain in engineering Voigt form (the three normal strains, then the three shears), satisfies one Lippmann–Schwinger equation [18]:

$$\underbrace{\begin{pmatrix}\mathbf{u}\\ \boldsymbol{\varepsilon}\end{pmatrix}(\mathbf{x})}_{\psi,\ 3+6} = \begin{pmatrix}\mathbf{u}^0\\ \boldsymbol{\varepsilon}^0\end{pmatrix}(\mathbf{x}) + \int_V \underbrace{\begin{pmatrix}\mathscr{G}_{uu} & \mathscr{G}_{u\varepsilon}\\ \mathscr{G}_{\varepsilon u} & \mathscr{G}_{\varepsilon\varepsilon}\end{pmatrix}(\mathbf{x}-\mathbf{y})}_{\text{kernel},\ 9\times9} \underbrace{\begin{pmatrix}\omega^2\Delta\rho\,\mathbf{I}_3 & \mathbf{0}\\ \mathbf{0} & \Delta\mathscr{C}\end{pmatrix}}_{\Delta,\ 9\times9} \begin{pmatrix}\mathbf{u}\\ \boldsymbol{\varepsilon}\end{pmatrix}(\mathbf{y})\,\mathrm{d}^3y, \tag{3}$$

$$\Delta\mathscr{C} = \Delta\lambda\begin{pmatrix}\mathbf{1}\mathbf{1}^{\mathsf{T}} & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix} + 2\Delta\mu\,\mathbf{I}_6, \qquad \mathbf{1} = (1,1,1)^{\mathsf{T}}, \tag{4}$$

the Voigt form of δc in the convention of Nestor [18], with $2\Delta\mu$ on the whole diagonal; the factor $\frac{1}{2}$ that engineering shear needs is carried by the kernel. Figure 1 shows the structure. The contrast is block diagonal and turns the state into the sources it induces: a force $\omega^2\Delta\rho\,\mathbf{u}$ from the displacement and a moment $\Delta\mathscr{C}\boldsymbol{\varepsilon}$ from the strain. The kernel's four blocks are the background's response to them, read as displacement or as strain. The moment radiates through the receiver derivative of $\mathbf{G}$, with a plus sign (the last term of Eq. (2)), and the strain is the symmetrised receiver derivative of the displacement. With $\partial_c = \partial/\partial x_c$, a Voigt row index I standing for the pair (a,d) of strain components and a Voigt column index J for the pair (b,c) of moment components (equal indices for a normal component, unequal for a shear), the blocks are

$$\begin{aligned}
\mathscr{G}_{uu,ab} &= G_{ab}, & \mathscr{G}_{u\varepsilon,aJ} &= \tfrac{1}{2}\big(\partial_c G_{ab} + \partial_b G_{ac}\big),\\
\mathscr{G}_{\varepsilon u,Ib} &= \tfrac{\eta_I}{2}\big(\partial_d G_{ab} + \partial_a G_{db}\big), & \mathscr{G}_{\varepsilon\varepsilon,IJ} &= \tfrac{\eta_I}{4}\big(\partial_d\partial_c G_{ab} + \partial_d\partial_b G_{ac} + \partial_a\partial_c G_{db} + \partial_a\partial_b G_{dc}\big),
\end{aligned} \tag{5}$$

with $\eta_I = 1$ on a normal row and 2 on a shear row, which makes the strain the engineering strain; all are evaluated at $\mathbf{x}-\mathbf{y}$. They are 3×3, 3×6, 6×3 and 6×6, and carry zero, one, one and two derivatives of $\mathbf{G}$. Each derivative raises the singularity at $\mathbf{x}=\mathbf{y}$ by one power of $1/r$, from $1/r$ to $1/r^3$; the last is integrable only as a principal value with a local term, and it is why the self-interaction of a cell needs the care of §3.

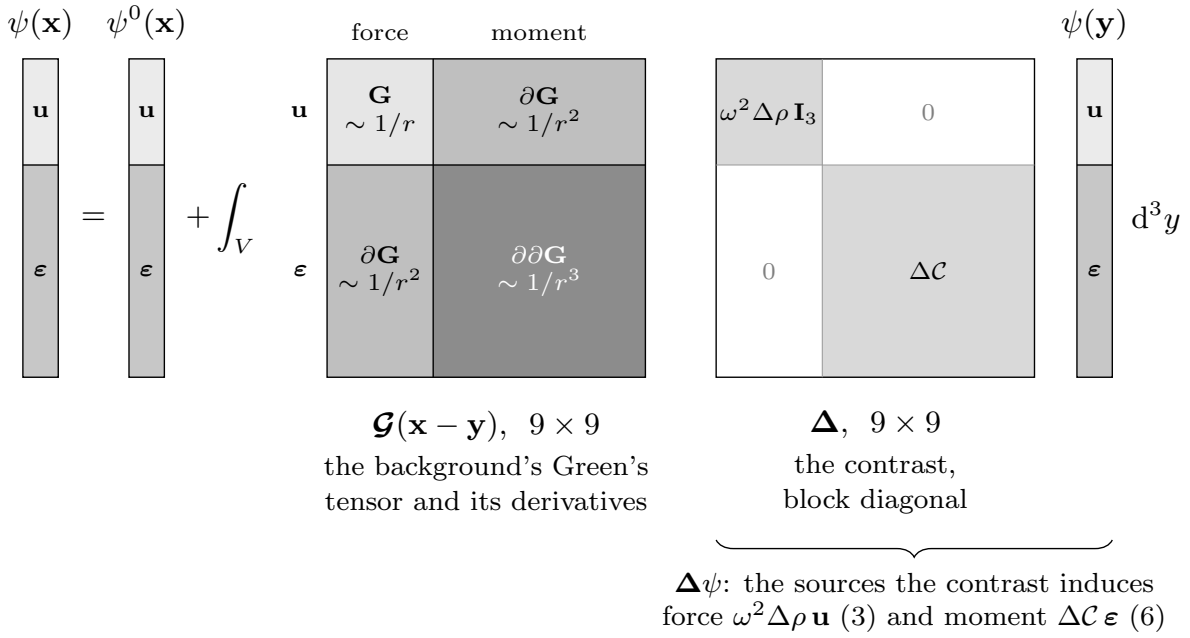


Figure 1. The nine-component Lippmann–Schwinger equation, Eq. (3), as block shapes; one unit is one component. The state ψ stacks the displacement (3) over the strain (6). The kernel's blocks carry zero, one, one and two derivatives of the Green's tensor, shaded by their singularity at $\mathbf{x} = \mathbf{y}$; its columns respond to a force and to a moment, its rows give displacement and strain. The contrast $\boldsymbol{\Delta}$ is block diagonal, so the density acts only through the displacement and the stiffness only through the strain.

The strain model. Table 1 sets out the three representations of the strain that this series meets. They differ in the unknowns, and the test of the equation follows from the unknowns: derivatives at a point are fixed by imposing the equation and its derivatives there, moments over a cell by testing against the cell's own polynomials. This paper is about the third.

Table 1. How the strain is represented within a cell, and the test that follows. The words *collocation* and *Galerkin* describe only the test.

strain model	unknowns of a cell	strain and displacement	test
uniform (the classical approximation)	displacement and strain at the centre	strain uniform	at the centre
derived	Legendre moments of displacement to degree p	strain the derivative of each cell's displacement polynomial, degree $p-1$	against the cell's polynomials (Galerkin)
independent (this paper)	Legendre moments of displacement and strain to degree p	independent, both degree p	against the cell's polynomials (Galerkin)

The derived strain loses most of the order: at 16 cells its errors are 3×10^4 times those of the independent strain for $p = 1$ and 5×10^8 times for $p = 2$ (`scripts/measure_derived_vs_independent_strain.py`). Restoring the jumps of the cellwise displacement across the faces does not recover it, so the loss is not the faces' alone; with the strain an unknown of its own, tested against the same polynomials as the displacement, the scheme reaches the order $2p + 2$ that a Galerkin scheme gives a functional of its solution [18].

Table 2. Derived against independent strain, on a layer of thickness D with an exact solution (P wave at normal incidence, $k_P D = 0.6$, $\Delta(\lambda + 2\mu) = 4\,\mathrm{GPa}$, $\Delta\rho = 100\,\mathrm{kg\,m^{-3}}$): relative error of the reflected displacement with 16 and 32 cells, and the order between them. All three schemes use Legendre polynomials of degree p in each cell, tested against the same. *Independent*: displacement and strain each of degree p. *Derived*: displacement of degree p, strain its derivative in each cell. *Derived, with faces*: the same, with the jumps of the displacement across the faces between cells added to the stiffness source as plane deltas, which makes the derivative that of a distribution.

	independent			derived			derived, with faces		
p	$n = 16$	32	order	16	32	order	16	32	order
1	2.6×10^{-9}	1.6×10^{-10}	4.00	9.0×10^{-5}	2.2×10^{-5}	2.00	1.4×10^{-4}	3.6×10^{-5}	1.98
2	2.6×10^{-14}	round-off	6.00 [a]	1.4×10^{-5}	3.5×10^{-6}	2.00	7.7×10^{-7}	9.5×10^{-8}	3.01
3	round-off	round-off	7.96 [b]	8.2×10^{-10}	5.1×10^{-11}	4.00	1.4×10^{-9}	8.8×10^{-11}	3.98

[a] from 4 to 8 cells. [b] from 2 to 4 cells; finer grids are at round-off.

The degree-one Legendre cell. A cube $[-h, h]^3$ with local coordinates $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_c)/h$ carries $\psi(\mathbf{x}) = \sum_b \psi_b L_b(\boldsymbol{\xi})$ over the four functions $L_b \in \{1, \xi_0, \xi_1, \xi_2\}$, each ψ_b a nine-component vector: 36 unknowns. With the contrast uniform in the cell the source density $\boldsymbol{\Delta}\psi$ is a polynomial on the same four functions. Testing Eq. (2) and its derivative against each L_a gives the single-site equation [17]

$$(\mathbf{M} - \mathbf{K}\boldsymbol{\Delta})\psi = \langle L, \psi^0 \rangle, \qquad \mathbf{M}_{ab} = \langle L_a, L_b \rangle \mathbf{I}_9, \tag{6}$$

with $\mathbf{M} = \mathrm{diag}(8h^3, \frac{8}{3}h^3, \frac{8}{3}h^3, \frac{8}{3}h^3) \otimes \mathbf{I}_9$ and $\mathbf{K}$ the self block, the kernel integrated against L_a over the cell as receiver and L_c over the cell as source. The single-site matrix maps the Legendre coefficients of the incident field to the source moments $\sigma_a = \langle L_a, \boldsymbol{\Delta}\psi \rangle$, the mean force and stress and their first moments,

$$\mathbf{T}_{36} = \mathbf{M}\boldsymbol{\Delta}\big(\mathbf{M} - \mathbf{K}\boldsymbol{\Delta}\big)^{-1}\mathbf{M} = \mathbf{M}\boldsymbol{\Delta}\big(\mathbf{I} - \mathbf{N}\boldsymbol{\Delta}\big)^{-1}, \qquad \mathbf{N} = \mathbf{M}^{-1}\mathbf{K}. \tag{7}$$

Everything specific to the cube is in $\mathbf{N}$; the contrast enters only through $\boldsymbol{\Delta}$.

3 The self block as a series in the wavenumber

The series. The Green's tensor is $\mathbf{G} = (4\pi\mu)^{-1}[\delta_{ij} g_S + k_S^{-2} \partial_i \partial_j (g_S - g_P)]$, with $k_P = \omega/\alpha$ and $k_S = \omega/\beta$ the wavenumbers of the background's compressional and shear waves, speeds α and β, and $g_k = \mathrm{e}^{\mathrm{i}kr}/r = \sum_t (\mathrm{i}k)^t r^{t-1}/t!$, so it is a sum over the powers r^m, $m \geq -1$, of two families, $\delta_{ij} r^m$ and $\partial_i \partial_j r^m$ [17]. The propagator adds up to two derivatives. Each entry of the self block is therefore a combination, with 9×9 tables TA_ι and TB_ι fixed by the propagator's structure, of the universal moments of the self cell

$$U_{ac}(m, \iota) = \int W_{ac}(\mathbf{s})\, \partial^\iota r^m \big|_{\mathbf{s}}\, \mathrm{d}^3 s, \qquad h = 1, \tag{8}$$

W_{ac} the cross-correlation of L_a and L_c over the cell, $\mathbf{s}$ the separation of a receiving and a radiating point, and $\boldsymbol{\iota}$ a multi-index of at most four derivatives. Collecting the powers of the wavenumber,

$$\mathbf{K} = \sum_{n\geq 0} (k_S h)^n \mathbf{K}_n, \qquad 4\pi\mu\,\mathbf{K}_n = \mathrm{i}^n\Big[\frac{\mathbf{A}_n}{n!} - \big(1-\gamma^{n+2}\big)\frac{\mathbf{B}_n}{(n+2)!}\Big], \qquad \gamma = \frac{k_P}{k_S} = \frac{\beta}{\alpha}, \tag{9}$$

$$\mathbf{A}_n = \sum_{\boldsymbol{\iota}} \mathsf{TA}_{\boldsymbol{\iota}}\, h^{5-|\boldsymbol{\iota}|}\, U(n-1,\boldsymbol{\iota}), \qquad \mathbf{B}_n = \sum_{\boldsymbol{\iota}} \mathsf{TB}_{\boldsymbol{\iota}}\, h^{7-|\boldsymbol{\iota}|}\, U(n+1,\boldsymbol{\iota}). \tag{10}$$

The term $n = 0$ is the static (Kelvin) block. The series is that of $\mathrm{e}^{\mathrm{i}kr}$ and converges for every k; at $k_S h \leq 0.2$ seven terms leave less than 4×10^{-9} of the T-matrix (§6). An entry of row type r and column type c (0 for displacement or force, 1 for strain or moment) scales as h^{5-r-c}, so the coefficients need be computed once, at $h = 1$.

The moments, exactly. For odd m the kernel r^m is singular or not smooth at $\mathbf{s} = 0$, which for the self cell is the centre of the support. The integral of Eq. (8) is then reduced, exactly as for two cells that touch [17]: the derivatives that make the kernel a distribution are moved onto W, whose second derivative carries plane deltas at its kinks; W is a polynomial on each of the eight boxes $[0,\pm2]^3$ that meet at the singular point; and each piece is a box or face integral of a monomial times a power of the distance, reduced from volume to faces to edges by recurrences whose base cases are elementary. Carried out in exact arithmetic, every such moment is a closed form. For even m the kernel is a polynomial, $\partial^{\boldsymbol{\iota}} r^m = 0$ for $|\boldsymbol{\iota}| > m$, and the box integrals are rational. Through $(k_S h)^6$ the self block needs 223 moments, $m = -1$ to 7; each is derived symbolically (`scripts/derive_legendre_cell_dynamic.py`).

4 Six constants, and a rational radiation

The moments of Eq. (8) for odd m are built from the master integrals of the self cell's boxes, whose sides are 2 and whose corners lie at 0 and ±2. Their recurrences only multiply powers of $\sqrt{a^2+b^2+c^2}$ with $a,b,c \in \{0,2\}$, which gives $\sqrt{2}$ and $\sqrt{3}$, and end in three base cases: $\operatorname{arsinh}(c/\sqrt{a^2+b^2})$, which gives $\operatorname{arsinh}1 = \ln(1+\sqrt{2})$ and

$$\operatorname{arsinh}\frac{1}{\sqrt{2}} = \ln\frac{1+\sqrt{3}}{\sqrt{2}} = \tfrac{1}{2}\ln\big(2+\sqrt{3}\big), \tag{11}$$

since $(1+\sqrt{3})^2/2 = 2+\sqrt{3}$, and $\arctan\big(bc/a\sqrt{a^2+b^2+c^2}\big) = \pi/6$. So for every odd m, and for every derivative, $U(m,\boldsymbol{\iota})$ is a combination with rational coefficients of

$$1, \qquad \pi, \qquad \sqrt{2}, \qquad \sqrt{3}, \qquad \ln\big(1+\sqrt{2}\big), \qquad \ln\big(2+\sqrt{3}\big). \tag{12}$$

The two logarithms come from the geometry of the cube: $\ln(1+\sqrt{2}) = \operatorname{arsinh}1$ from its faces, whose diagonal is $\sqrt{2}$ times the side, and $\ln(2+\sqrt{3}) = 2\operatorname{arsinh}(1/\sqrt{2})$ from the cube itself, whose body diagonal is $\sqrt{3}$ times the side. It appears elsewhere as $-\ln(2-\sqrt{3})$, the same number. We write it as $\ln(2+\sqrt{3})$, which avoids the cancellation in $2-\sqrt{3}$ and is the better conditioned in floating point. By Eq. (10), $\mathbf{K}_n$ draws on $m = n-1$ and $n+1$. An even power of $k_S h$ therefore uses

odd m and is a combination of the six constants; an odd power uses even m, polynomial kernels, and is rational. Divided by the 4π of Eq. (9), an even power is a rational combination of $1/\pi$, 1, $\sqrt{2}/\pi$, $\sqrt{3}/\pi$, $\ln(1+\sqrt{2})/\pi$ and $\ln(2+\sqrt{3})/\pi$, and an odd power a rational multiple of $1/\pi$. The derivation confirms both through the highest power computed: the constants that occur in $\mathbf{K}_0$, $\mathbf{K}_2$, $\mathbf{K}_4$ and $\mathbf{K}_6$ are exactly the six, and every entry of $\mathbf{K}_1$, $\mathbf{K}_3$ and $\mathbf{K}_5$ is rational. In the static strain blocks the two logarithms occur only in the combination $\Lambda = [\ln(1+\sqrt{2}) - \frac{1}{2}\ln(2+\sqrt{3})]/\pi$ of Nestor [17]; in the displacement blocks and at the dynamic orders they occur separately.

The radiation. The odd powers are the radiation of the cell. The first, $n = 1$, has a single non-zero entry: the uniform displacement against the mean force, the cell's monopole. In the reduced form of §5 it is

$$N_{T_{1u},1}[u,u] = \frac{\mathrm{i}}{\mu}\Big[\frac{2}{\pi} - \big(1-\gamma^3\big)\frac{4}{3!\,\pi}\Big] = \frac{2\mathrm{i}\,(2+\gamma^3)}{3\pi\mu}, \tag{13}$$

so that the term $k_S h\, h^2 N_{T_{1u},1}$ of $\mathbf{M}^{-1}\mathbf{K}$ is $8h^3$ times $\mathrm{i}\,k_S(2+\gamma^3)/(12\pi\mu)$, the imaginary part of $\mathbf{G}(0)$: the radiation damping of a point force carrying the cell's total force, as it must be at lowest order. The next odd power, $n = 3$, is again rational; it enters A_{1g}, E_g, T_{1g}, T_{2g} and T_{1u}, and vanishes on T_{2u}, A_{2u} and E_u; the fifth enters every representation.

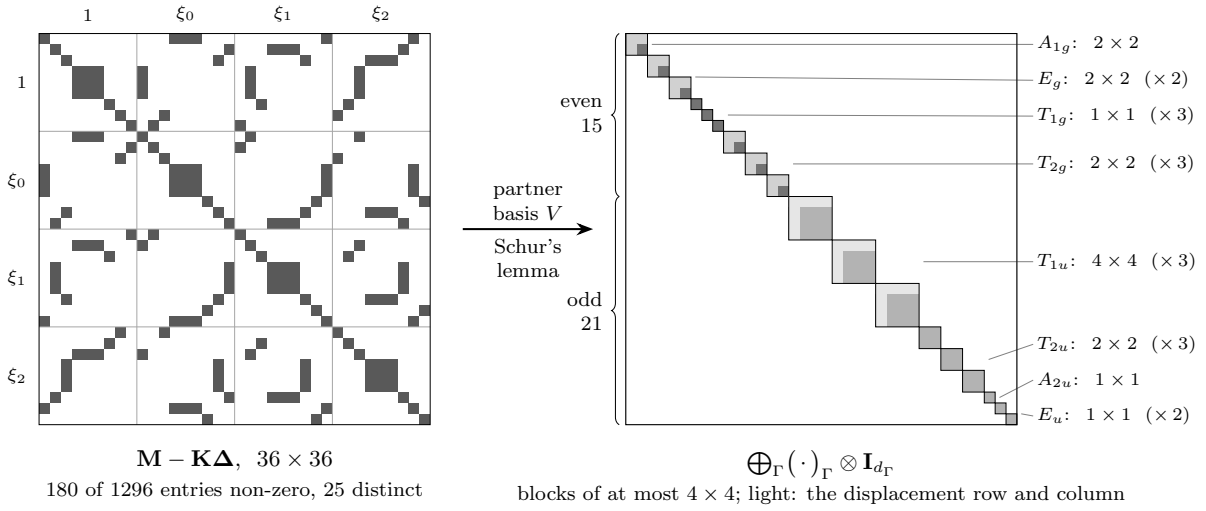
5 The cube group

With the contrast uniform in the cell, the 48 symmetries of the cube (the signed permutations of its axes, the group O_h) act on the 36 unknowns, and the self block, the Gram matrix and the contrast all commute with that action: the cube is mapped onto itself, the isotropic Green's tensor is covariant, and the uniform contrast is invariant. By Schur's lemma the single-site system then splits exactly into independent blocks, one for each irreducible representation Γ of O_h that occurs,

$$\mathbf{K} = \bigoplus_{\Gamma} \big(\mathbf{K}_\Gamma \otimes \mathbf{I}_{d_\Gamma}\big), \tag{14}$$

an $n_\Gamma \times n_\Gamma$ block, n_Γ the number of times Γ occurs, repeated for each of the d_Γ rows of Γ; likewise for $\mathbf{M}$, $\mathbf{\Delta}$ and $\mathbf{T}_{36}$. The commutation holds for the complex self block at every frequency, radiation included, so the splitting holds power by power. The displacement's first moments $\xi_i u_j$ (9 unknowns) span $A_{1g} + E_g + T_{1g} + T_{2g}$; the uniform strain (6) spans $A_{1g} + E_g + T_{2g}$; the uniform displacement (3) spans T_{1u}; and the strain's first moments (18) span $3T_{1u} + 2T_{2u} + A_{2u} + E_u$. Altogether there are 15 even unknowns, $2A_{1g} + 2E_g + T_{1g} + 2T_{2g}$, and 21 odd, $4T_{1u} + 2T_{2u} + A_{2u} + E_u$; even and odd never couple, and the largest block is T_{1u}, 4×4. Written out, with V the matrix whose columns are the partner vectors (Appendix A),

$$\begin{aligned} V^{-1}\,\mathbf{T}_{36}\,V = {} & \underbrace{\mathbf{T}_{A_{1g}}}_{2\times2} \oplus \big(\underbrace{\mathbf{T}_{E_g}}_{2\times2} \otimes \mathbf{I}_2\big) \oplus \big(\underbrace{\mathbf{T}_{T_{1g}}}_{1\times1} \otimes \mathbf{I}_3\big) \oplus \big(\underbrace{\mathbf{T}_{T_{2g}}}_{2\times2} \otimes \mathbf{I}_3\big) \qquad \text{(even, 15)} \\ & \oplus \big(\underbrace{\mathbf{T}_{T_{1u}}}_{4\times4} \otimes \mathbf{I}_3\big) \oplus \big(\underbrace{\mathbf{T}_{T_{2u}}}_{2\times2} \otimes \mathbf{I}_3\big) \oplus \underbrace{\mathbf{T}_{A_{2u}}}_{1\times1} \oplus \big(\underbrace{\mathbf{T}_{E_u}}_{1\times1} \otimes \mathbf{I}_2\big) \quad \text{(odd, 21)}, \end{aligned} \tag{15}$$



$\mathbf{M} - \mathbf{K}\boldsymbol{\Delta}$, 36×36
180 of 1296 entries non-zero, 25 distinct

$\bigoplus_\Gamma (\,\cdot\,)_\Gamma \otimes \mathbf{I}_{d_\Gamma}$
blocks of at most 4×4; light: the displacement row and column

Figure 2. The single-site system before and after the change to the partner basis. Left: the non-zero entries of $\mathbf{M} - \mathbf{K}\boldsymbol{\Delta}$ (dark) in the natural ordering, the four functions $1, \xi_0, \xi_1, \xi_2$ each carrying the nine components, at $k_S h = 0.1$; 180 of the 1296 entries are non-zero, and they take only 25 distinct values up to sign, the symmetry of the cube repeating them within and between the blocks of the functions. Right: the same matrix in the basis of partner vectors (Table 6), by Schur's lemma block diagonal, Eq. (15): each irreducible representation gives one block, of size its multiplicity, repeated once for each of its rows; the 15 even unknowns (darker) never couple to the 21 odd (lighter). Within a block, the lighter row and column are its single displacement vector, coupled to the strain only through $\omega^2 \Delta\rho$ (Eq. (21)). The same structure holds at every frequency and for every power of $k_S h$, and for $\mathbf{T}_{36}$.

each block given by Eq. (16) below and $\mathbf{I}_d$ repeating it once for each row of its representation. The reduction does not make the matrix hold fewer numbers: in the natural ordering the 180 non-zero entries of $\mathbf{M} - \mathbf{K}\boldsymbol{\Delta}$ take only 25 distinct values up to sign, the symmetry of the cube repeating them within and between the blocks of the four functions, while the eight blocks hold 35 entries, combinations of those values in the partner basis. What it changes is the coupling: the 36 unknowns, coupled through those entries in one 36×36 system, separate into independent systems of at most four unknowns (Figure 2), each inverted on its own, which is what lets $\mathbf{T}_{36}$ be written out in closed form.

Appendix A gives the group's action on the unknowns, why the commutation holds, Schur's lemma and what it requires, and the vectors that span one row of each block (Table 6); the reduction is exact, with no residual at any power computed (§6).

On each representation

$$\mathbf{T}_\Gamma = M_\Gamma \boldsymbol{\Delta}_\Gamma \big(\mathbf{I} - \mathbf{N}_\Gamma \boldsymbol{\Delta}_\Gamma\big)^{-1}, \qquad \mathbf{N}_\Gamma = \frac{1}{\mu} \sum_{n \geq 0} (k_S h)^n \, \mathrm{i}^n \Big[\frac{\mathbf{P}_{\Gamma,n}}{n!} - \big(1 - \gamma^{n+2}\big) \frac{\mathbf{Q}_{\Gamma,n}}{(n+2)!} \Big], \tag{16}$$

with $\boldsymbol{\Delta}_\Gamma$ the contrast on the vectors of Γ, which is $\omega^2 \Delta\rho$ on a displacement vector and a combination of $\Delta\lambda$ and $\Delta\mu$ on a strain vector, and $\mathbf{P}_{\Gamma,n}$, $\mathbf{Q}_{\Gamma,n}$ the reduced forms of $\mathbf{A}_n$ and $\mathbf{B}_n$ divided by $4\pi M_\Gamma$. Each of their entries is a rational vector on the six constants (§4); all of them, through the highest power computed, are in `data/self_blocks_irreps.txt`. As an example, the uniform dilatation (A_{1g}, the strain vector) has $P_0 = -\frac{1}{3}$ and $Q_0 = -\frac{2}{3}$, so that its static entry is $\mu N = -\frac{1}{3} + \frac{1}{2}(1 - \gamma^2)\frac{2}{3} = -\gamma^2/3$, that is $N = -1/[3(\lambda + 2\mu)]$, the value for every shape [17]; all

36 static entries of the uniform and linear strain reproduce those of Nestor [17], coefficient by coefficient on its five constants. Its first dynamic correction is

$$\mu N_{A_{1g},2}[\varepsilon,\varepsilon] = -\frac{1}{2}\Big[P_2 - \frac{1-\gamma^4}{12}Q_2\Big], \qquad \pi P_2 = \tfrac{4}{15} - \tfrac{4}{9}\pi + \tfrac{4}{15}\sqrt{2} - \tfrac{8}{15}\sqrt{3} + \tfrac{4}{3}\ln(1+\sqrt{2}) + \tfrac{4}{3}\ln(2+\sqrt{3}), \tag{17}$$

and likewise for Q_2 (the data file); $(k_S h)^2$ times this is the leading change with frequency of the cell's dilatational self-interaction.

Inversion in closed form. The reduction matters because of what it does to the inverse in Eq. (7). On the irreducible representations the 36×36 matrix to be inverted becomes eight blocks of order at most four, and each block's inverse is an explicit formula in the contrast and in $k_S h$: the T-matrix is a closed-form function of the material and the frequency, not a number obtained for one contrast and one frequency at a time. The structure is not peculiar to the Galerkin cell. The author first found it, with a closed-form inverse and the physical reading that follows, for the cell with derived strain, the displacement expanded about the centre to its second gradients (§2), in an unpublished notebook [22]. There too the system splits by parity: the first gradients form a block of their own, while the displacement couples only to the second gradients; the displacement follows from a Schur complement, $\mathbf{u} = (\mathbf{A}_{11} - \mathbf{A}_{13}\mathbf{A}_{33}^{-1}\mathbf{A}_{31})^{-1}(\cdots)$, the counterpart of Eq. (21); and in the static limit the strain is $(\mathbf{I} - \mathsf{M}\,\delta c)^{-1}$ times the incident strain, with M the integral of the second derivatives of $\mathbf{G}$ over the cell: Eshelby's form. It is derived here for the Legendre cell, whose strain is an unknown of its own, at every power of $k_S h$, with each step checked. On each representation the contrast is the same simple matrix whatever the cell's material, from Table 6 and the contrast operator of §2:

$$\begin{gathered}
\boldsymbol{\Delta}_{T_{1g}} = \omega^2\Delta\rho, \qquad \boldsymbol{\Delta}_{A_{2u}} = \boldsymbol{\Delta}_{E_u} = 2\Delta\mu, \qquad \boldsymbol{\Delta}_{T_{2u}} = 2\Delta\mu\,\mathbf{I}_2, \\
\boldsymbol{\Delta}_{A_{1g}} = \operatorname{diag}\big(\omega^2\Delta\rho,\ 3\Delta\lambda + 2\Delta\mu\big), \qquad \boldsymbol{\Delta}_{E_g} = \boldsymbol{\Delta}_{T_{2g}} = \operatorname{diag}\big(\omega^2\Delta\rho,\ 2\Delta\mu\big), \\
\boldsymbol{\Delta}_{T_{1u}} = \omega^2\Delta\rho \oplus \begin{pmatrix} \Delta\lambda + 2\Delta\mu & 2\Delta\lambda \\ \Delta\lambda & 2\Delta\lambda + 2\Delta\mu \end{pmatrix} \oplus 2\Delta\mu.
\end{gathered} \tag{18}$$

With $\mathbf{X}_\Gamma = \mathbf{N}_\Gamma\boldsymbol{\Delta}_\Gamma$, Eq. (16) is inverted by Cramer's rule,

$$\begin{gathered}
\mathbf{T}_\Gamma = \frac{M_\Gamma \boldsymbol{\Delta}_\Gamma \operatorname{adj}(\mathbf{I} - \mathbf{X}_\Gamma)}{\det(\mathbf{I} - \mathbf{X}_\Gamma)}, \qquad \text{for instance} \\
(\mathbf{I} - \mathbf{X})^{-1} = \frac{1}{(1-X_{11})(1-X_{22}) - X_{12}X_{21}} \begin{pmatrix} 1 - X_{22} & X_{12} \\ X_{21} & 1 - X_{11} \end{pmatrix}
\end{gathered} \tag{19}$$

for a block of order two, and $M\Delta/(1 - N\Delta)$ for a block of order one; the block of order four, T_{1u}, has a determinant of degree four in the contrast. Three things follow, none of which a numerical inverse gives.

The T-matrix is an explicit function of the contrast. Each entry of $\mathbf{T}_\Gamma$ is a ratio of polynomials in $\Delta\lambda$, $\Delta\mu$ and $\omega^2\Delta\rho$, of degree at most n_Γ, with coefficients the entries of $\mathbf{N}_\Gamma$, which carry the frequency through Eq. (16) and nothing else of the material. A new contrast needs no new solve, and the derivatives of $\mathbf{T}$ with respect to the contrast, which an inversion for the medium needs,

are those of explicit rational functions.

Its poles are written down. $\mathbf{T}_\Gamma$ is singular where $\det(\mathbf{I} - \mathbf{N}_\Gamma\boldsymbol{\Delta}_\Gamma) = 0$, one condition per representation, explicit in the contrast; and the expansion of $(\mathbf{I} - \mathbf{X}_\Gamma)^{-1}$ in powers of the contrast, the Born series of the cell, converges exactly when the eigenvalues of the at most 4×4 matrix $\mathbf{X}_\Gamma$ lie inside the unit circle.

Classical results appear as special cases. In the static limit the uniform dilatation decouples ($\omega^2\Delta\rho \to 0$ removes the displacement row of A_{1g}), $N = -1/[3(\lambda+2\mu)]$, and with $D = 3\Delta\lambda + 2\Delta\mu = 3\Delta K$, $K = \lambda + \frac{2}{3}\mu$ the bulk modulus and $K' = K + \Delta K$,

$$T_{A_{1g}} = \frac{8h^3 D}{1 + D/[3(\lambda + 2\mu)]} = 8h^3\, 3\Delta K\, \frac{K + \frac{4}{3}\mu}{K' + \frac{4}{3}\mu}, \tag{20}$$

Eshelby's dilatational result for a sphere [19], with its pole at $K' = -\frac{4}{3}\mu$; the closed form agrees with the numerical $\mathbf{T}_{36}$ to 1.1×10^{-15} (§6). The two shear channels E_g and T_{2g} take the same form with the cube's constants of Nestor [17] in place of the sphere's.

Each block is simpler still. Every block holds at most one displacement vector (Table 6), and the contrast on it is $\omega^2\Delta\rho = (k_S h)^2\beta^2\Delta\rho/h^2$, which vanishes in the static limit. Ordering a block as its displacement vector u and its strain vectors ε, the inverse follows from the Schur complement on that single unknown:

$$(\mathbf{I} - \mathbf{X})^{-1} = \begin{pmatrix} \dfrac{1}{a} + \dfrac{\mathbf{b}\,\mathbf{S}^{-1}\mathbf{c}}{a^2} & -\dfrac{\mathbf{b}\,\mathbf{S}^{-1}}{a} \\ -\dfrac{\mathbf{S}^{-1}\mathbf{c}}{a} & \mathbf{S}^{-1} \end{pmatrix}, \qquad \begin{aligned} a &= 1 - X_{uu}, \quad \mathbf{b} = -\mathbf{X}_{u\varepsilon}, \quad \mathbf{c} = -\mathbf{X}_{\varepsilon u}, \\ \mathbf{S} &= \mathbf{I} - \mathbf{X}_{\varepsilon\varepsilon} - \mathbf{c}\,\mathbf{b}/a, \end{aligned} \tag{21}$$

where a is a number and $\mathbf{X}_{\varepsilon u} = \mathbf{N}_{\varepsilon u}\,\omega^2\Delta\rho$, the only route by which the displacement reaches the strain, is of order $(k_S h)^2$. What remains to invert is the strain matrix $\mathbf{S}$: a number for A_{1g}, E_g, T_{2g}, A_{2u} and E_u, a 2×2 matrix for T_{2u} and a 3×3 one for T_{1u}, while T_{1g} is the displacement alone (Table 3).

Table 3. What each block leaves to invert. A block of size $n_\Gamma \times n_\Gamma$ is repeated d_Γ times (Eq. (15)); $\boldsymbol{\Delta}_\Gamma$ is its contrast, Eq. (18); $a = 1 - X_{uu}$ is the number of Eq. (21) for its displacement vector, and $\mathbf{S}$ the strain matrix. In the static limit ($k_S h \to 0$, $\omega^2\Delta\rho \to 0$) every displacement drops out.

Γ	block ($\times d_\Gamma$)	contrast $\boldsymbol{\Delta}_\Gamma$	left to invert	static limit
A_{1g}	2×2	$\mathrm{diag}(\omega^2\Delta\rho,\ 3\Delta\lambda + 2\Delta\mu)$	a; $\mathbf{S}$ a number	Eq. (22), Eshelby's sphere
E_g	2×2 ($\times 2$)	$\mathrm{diag}(\omega^2\Delta\rho,\ 2\Delta\mu)$	a; $\mathbf{S}$ a number	Eq. (22)
T_{1g}	1×1 ($\times 3$)	$\omega^2\Delta\rho$	a alone	vanishes (rotation)
T_{2g}	2×2 ($\times 3$)	$\mathrm{diag}(\omega^2\Delta\rho,\ 2\Delta\mu)$	a; $\mathbf{S}$ a number	Eq. (22)
T_{1u}	4×4 ($\times 3$)	$\omega^2\Delta\rho \oplus (2 \times 2) \oplus 2\Delta\mu$	a; $\mathbf{S}$ 3×3	the 3×3 strain block
T_{2u}	2×2 ($\times 3$)	$2\Delta\mu\,\mathbf{I}_2$	2×2: two scalar channels	two channels of Eq. (22)'s form
A_{2u}	1×1	$2\Delta\mu$	a number	Eq. (22)
E_u	1×1 ($\times 2$)	$2\Delta\mu$	a number	Eq. (22)

In the static limit $\mathbf{c} = 0$, the displacement drops out, and the five uniform-strain and scalar

channels are each of the form of Eq. (20),

$$T_\Gamma = \frac{M_\Gamma D_\Gamma}{1 - N_\Gamma D_\Gamma}, \qquad D_\Gamma = 3\Delta\lambda + 2\Delta\mu \ (A_{1g}), \quad 2\Delta\mu \ (E_g,\ T_{2g},\ A_{2u},\ E_u), \tag{22}$$

one number N_Γ per channel. For T_{2u} the contrast is a multiple of the identity, $2\Delta\mu\,\mathbf{I}_2$, so the two eigenvectors of $\mathbf{N}_{T_{2u}}$ split it into two channels of the same form. Only the strain's first moments in T_{1u}, three unknowns coupled by the 2×2 normal-stress part of the contrast, need a 3×3 inverse. The structure is exact: the displacement column of $\mathbf{X}$ vanishes to round-off as $k_S h \to 0$, and Eq. (21) reproduces the numerical inverse of every block to 4.4×10^{-16} for $k_S h$ up to 0.3.

The static cell is a set of Eshelby channels. Put together, the static single-site T-matrix of the cube is, apart from the three coupled strain moments of T_{1u}, a set of independent scalar channels,

$$\boxed{T_\Gamma = \frac{M_\Gamma D_\Gamma}{1 - N_\Gamma D_\Gamma}} \tag{23}$$

each with the contrast D_Γ that acts on it and one constant N_Γ, the cube's counterpart of Eshelby's tensor for that channel (Table 4). This is the form Eshelby found for an ellipsoid [19], where the self-interaction of a uniform eigenstrain is uniform; the cube's is not, and it has more channels, but each keeps the form, with its constant in closed form. For the dilatation the constant is the sphere's, as it is for every shape. The strain's first moments add the channels A_{2u}, E_u and the two of T_{2u}, the cube's counterparts of the linear eigenstrains of an ellipsoid [20].

The cube's two shear channels average to the sphere's. A sphere has one uniform-shear channel, five-dimensional. The cube splits it into E_g (two dimensions) and T_{2g} (three), with different constants, -0.2268 and -0.2622 in units of $1/\mu$. Weighted by their dimensions they average exactly to the sphere's constant:

$$\boxed{\tfrac{1}{5}\left(2N_{E_g} + 3N_{T_{2g}}\right) = -\frac{4 - 5\nu}{15\mu(1 - \nu)}} \tag{24}$$

for every Poisson ratio. From Table 4, $\mu(2N_{E_g} + 3N_{T_{2g}}) = -\frac{5}{3} + b_2\left(6\chi_1 - \frac{4}{3} - 6\chi_1\right) = -\frac{5}{3} - \frac{4}{3}b_2$: the cube's constant χ_1 cancels, and with $1 - \beta^2/\alpha^2 = 1/[2(1 - \nu)]$ what remains is the sphere's value. At $\nu = 7/32$ both sides are -0.2480. The reason is that the sum over all five shear directions is the trace of the operator on the deviatoric strains, and that trace sees only the isotropic part of the self-interaction, which is the same for every shape of the same volume; the cube's anisotropy only moves the constant from one shear channel to the other. The derivation script verifies Eq. (24) symbolically, with β/α left free (Table 5).

6 Checks

Every step is checked against an evaluation that shares none of it (`scripts/derive_legendre_cell_dynamic.py`), with the background $\alpha = 5000\,\mathrm{m\,s^{-1}}$, $\beta = 3000\,\mathrm{m\,s^{-1}}$, $\rho = 2500\,\mathrm{kg\,m^{-3}}$ and

Table 4. The static channels of the degree-one Legendre cell, Eq. (23). N_Γ is the static entry of $\mathbf{N}_\Gamma$ on the channel, $\mu N_\Gamma = p + b_2 q$ with $b_2 = -\frac{1}{2}(1 - \beta^2/\alpha^2)$, on the constants χ_1, χ_0 and Λ of Nestor [17]; the values are for $\beta/\alpha = 3/5$ ($\nu = 7/32$). The q of A_{2u}, E_u and T_{2u} is in the data file of the Data availability section. The sphere's shear constant at this ν is $\mu N = -(4 - 5\nu)/[15(1 - \nu)] = -0.2480$.

Γ	unknowns	D_Γ	μN_Γ	value
A_{1g}	uniform dilatation	$3\Delta\lambda + 2\Delta\mu$	$-\frac{1}{3} - \frac{2}{3}b_2 = -\frac{1}{3}\beta^2/\alpha^2$	-0.1200
E_g	uniform deviatoric normal strain	$2\Delta\mu$	$-\frac{1}{3} + b_2(3\chi_1 - \frac{2}{3})$	-0.2268
T_{2g}	uniform shear	$2\Delta\mu$	$-\frac{1}{3} - 2b_2\chi_1$	-0.2622
A_{2u}	first moments of shear	$2\Delta\mu$	$-\frac{1}{2} + 6\chi_0 + b_2 q$	-0.2135
E_u	first moments of shear	$2\Delta\mu$	$-\frac{1}{4} + 3\chi_0 + b_2 q$	-0.1296
T_{2u}	first moments, two channels	$2\Delta\mu$	eigenvalues of $\mu\mathbf{N}_{T_{2u}}$	-0.1183, -0.3097

the contrast $\Delta\lambda = 2\,\mathrm{GPa}$, $\Delta\mu = 1\,\mathrm{GPa}$, $\Delta\rho = 100\,\mathrm{kg\,m^{-3}}$. Table 5 gives the results.

Table 5. Checks of the closed form through $(k_S h)^6$. The reference for the series is the package's series of the self block (master integrals at forty digits for odd m, Gauss rules for even m), which itself agrees with quadrature of the singular integrals to 1.4×10^{-15}.

check	result
223 exact moments, $m = -1$ to 7, against the package's	1.7×10^{-15}
self block, exact series to $n = 6$, against the full series, $k_S h = 0.02, 0.05, 0.1, 0.2$	1.1×10^{-14}, 6.6×10^{-12}, 8.4×10^{-10}, 1.1×10^{-7}
apparent order of that difference	7.00
$\mathbf{T}_{36}$ from the exact series against $\mathbf{T}_{36}$ by quadrature, $k_S h = 0.05, 0.1, 0.2$	2.1×10^{-13}, 2.7×10^{-11}, 3.5×10^{-9}
apparent order	7.00
the same truncated at $n = 4$	5.9×10^{-10}, 1.9×10^{-8}, 6.1×10^{-7}; order 5.00
invariance of every representation's span under every $\mathbf{K}_n$	exact (no residual)
$\mathbf{T}_{36} V_\Gamma = V_\Gamma \mathbf{T}_\Gamma$, Eq. (16), worst over Γ, $k_S h = 0.05, 0.1, 0.2$	4.4×10^{-13}, 5.6×10^{-11}, 7.2×10^{-9}
static uniform dilatation	$-1/[3(\lambda + 2\mu)]$ exactly
static dilatational block of $\mathbf{T}_{36}$ against Eshelby's sphere, Eq. (20)	1.1×10^{-15}
Schur-complement inverse of every block, Eq. (21), $k_S h \le 0.3$	4.4×10^{-16}
dimension-weighted mean of the static E_g and T_{2g} constants against the sphere's shear constant, Eq. (24), β/α free	exact (no residual)
the 36 static strain entries against Nestor [17]	identical coefficients
radiation at $n = 1$	$\mathrm{Im}\,\mathbf{G}(0)$ exactly, Eq. (13)

The differences fall as $(k_S h)^7$, the first power omitted (and as $(k_S h)^5$ when the series stops at $n = 4$), so they are the truncation of the series and not an error of its terms. The closed form is that of the exact self block: what it omits at a given order is stated by the next term, which is itself in closed form.

7 Discussion

A frequency sweep with no integral at any frequency. The single-site T-matrix of the degree-one Legendre cell is now known in closed form at every power of the frequency, and what

that buys is a sweep. The coefficients $\mathbf{K}_n$ depend on neither the frequency, the cell size nor the medium: they are computed once, in exact arithmetic, and stored. Each frequency then takes two steps: evaluate each block $\mathbf{N}_\Gamma$ as a polynomial in $k_S h$ with stored coefficients (Eq. (9) reduced to the representation); then evaluate each block's T-matrix from its closed-form inverse, Eqs. (16) and (21). Nothing singular is integrated and nothing larger than 4×4 is inverted at any frequency. Measured on one machine (`scripts/measure_legendre_cell_sweep.py`), one frequency costs 0.43 ms by the eight block formulas, each the Schur complement on its displacement unknown with the complement inverted by Cramer's rule, against 1.2 s by quadrature of the singular self integrals, a factor of about 2800, and against 9.4 ms by the package's numerical series of the same moments. The one-off cost of the closed form, the 223 exact moments and their assembly and reduction through $(k_S h)^6$, is 30 s on four cores, paid once for every cell, frequency and medium; against quadrature it is recovered after about twenty-five frequencies, and a sweep of a thousand frequencies that quadrature would take twenty minutes to tabulate costs half a minute in all. The coupling between cells has the same structure [17]: its coefficients too are universal and computed once, and over 1024 frequencies its series costs 1.8 to 10.5 min against 34 to 98 min by quadrature on the grids measured there. With both, a model of cells is assembled at any frequency from stored numbers.

What the closed-form inverse buys. The block formulas are not the fastest way to a number: assembling $\mathbf{K}$ and handing the 36×36 system to a library solver takes 0.14 ms, a third of their cost, because at this size the time goes to the interpreter, not the arithmetic. They cost more code as well: the partner vectors of Table 6, the reduction of each $\mathbf{K}_n$, and one inverse formula per representation. What that code buys is that $\mathbf{T}$ stops being a number and becomes a function, with three consequences that a solver cannot supply (§5).

- *The material enters last and cheaply.* $\mathbf{N}_\Gamma$ depends only on the background and the frequency. In an inversion for the medium, where the contrast changes at every iteration and the background and frequencies do not, $\mathbf{N}_\Gamma$ is evaluated once per frequency and every later $\mathbf{T}$ is a handful of scalar operations on the new contrast; its derivatives with respect to $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$, the cell's sensitivities, are those of explicit rational functions, with no finite differences and no adjoint solve.

- *Each channel says what it sees.* The uniform dilatation (A_{1g}) responds to the bulk contrast only; E_g, T_{2g}, A_{2u}, E_u and T_{2u} to the shear contrast only; T_{1u} alone mixes the two. The density enters only the blocks that hold a displacement unknown, and there through the single number a of Eq. (21). Which elastic parameter is visible in which scattering channel is read from the formula, not inferred from computed patterns.

- *Resonance and convergence are written down.* Each block's poles are the zeros of $\det(\mathbf{I}-\mathbf{X}_\Gamma) = a\det\mathbf{S}$, explicit in the contrast and the frequency, and the cell's Born series converges exactly when the eigenvalues of a matrix of order at most four lie inside the unit circle. Near a resonance the ill-conditioning sits in these scalar denominators, where it can be monitored, rather than spread through a 36×36 solve.

The range of the truncated series. The series converges for every $k_S h$, but a sweep evaluates it truncated. The figures that follow are for the single-site T-matrix of one cell, compared entry by entry (largest difference relative to the largest entry) with the same T-matrix built on the self block by quadrature. The cell's interaction with its own field is summed exactly by the inverse in Eq. (7), and no other cell is involved: the truncation is only that of the series in the frequency of the self block. Through $(k_S h)^6$ the T-matrix is accurate to 3.5×10^{-9} at $k_S h = 0.2$, 5.9×10^{-8} at 0.3, 2.1×10^{-6} at 0.5 and 2.6×10^{-4} at 1, against quadrature that agrees with the package's series to 10^{-16}; stopping at $(k_S h)^4$ instead loses three orders of magnitude at $k_S h = 0.05$ and less than one at $k_S h = 1$. A discretisation with ten cells to the shear wavelength has $k_S h = \pi/10 \approx 0.31$, where the truncation through $(k_S h)^6$ leaves 8.2×10^{-8}. Beyond that, the next powers are in closed form like the first, from the same derivation, at a one-off cost that grows with the number of moments.

Eshelby's form, extended to the cube. The static cell is a set of independent channels of Eshelby's form, Eq. (23), with one closed-form constant each (Table 4). The static self-interaction of a cuboid inclusion is known in closed form point by point [23, 24]; it is not uniform, so a uniform-strain closure for the cube depends on where it is imposed. The Galerkin cell imposes it on average over the cell, and its constants are those averages, exact, channel by channel; the strain's first moments add channels of the same form. The dynamic T-matrix keeps the structure, with the displacement entering each channel through the single number a of Eq. (21). The closed-form inverse makes the physics visible, as it did for the derived-strain cell in the author's earlier notebook [22], which has the same block structure. The Legendre cell scatters through a few independent channels, each with one constant and one resonance, $1 - N_\Gamma D_\Gamma = 0$.

What is not yet covered. Two things. The contrast here is uniform in the cell; a graded contrast couples the even and odd unknowns and leaves only the operations that fix its gradient [17], so the reduction is weaker, though the moments are the same and the closed form carries over. And the cell is of degree one; the degree-two cell, whose expansions add the six quadratic Legendre polynomials, $P_2(\xi_i)$ and $\xi_i \xi_j$ $(i < j)$, to the four of degree one (10 functions, 90 unknowns), needs the moments of monomials to degree four, which the same derivation provides.

Data availability

The derivation and its checks are `scripts/derive_legendre_cell_dynamic.py`, which writes `data/self_moments_exact.pkl` (every moment, exact) and `data/self_blocks_irreps.txt` (every reduced entry, on the six constants), in the repository https://github.com/tmnestor/MultipleScatteringCalculations. The cost of a sweep and the range of the truncated series are measured by `scripts/measure_legendre_cell_sweep.py`. The static blocks are `scripts/derive_graded_voxel_site_blocks.py`.

Acknowledgments

This work began thirty years ago in the author's doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [21].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Software development. Software development followed a human-directed, AI-assisted workflow, organised as a Test Driven Hypothesis Acceptance cycle: each derivation, implementation or numerical claim was stated as a hypothesis together with a test designed to refute it (an independent route, an exact limit, a high-precision reference or a convergence study), and was accepted only when the test passed; a failed test returned the hypothesis for diagnosis, never the test for adjustment. The author designed the software originally, but only directed and validated recent improvements, which build on the author's original Fortran implementation (the `PhD_fortran_code` folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author's. All AI-generated changes were subject to code review by the author and validated by automated tests and independent checks: exact arithmetic, the symmetry of the cube, Eshelby's limits and convergence.

Manuscript preparation. During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

A The cube group and Schur's lemma

The splitting of §5 rests on Schur's lemma from the representation theory of finite groups [25, 26]. This appendix sets out the group's action on the unknowns of the cell, why the cell's matrices commute with it, and the form the lemma then gives them.

Symmetries of the cube. A *symmetry* of the cube is a linear map $\boldsymbol{\xi} \mapsto Q\boldsymbol{\xi}$ of space that preserves distances and maps the cube $[-1,1]^3$ onto itself; it fixes the centre, and Q is an orthogonal matrix, $Q^{\mathsf{T}}Q = \mathbf{I}$. Such a map must send the six face centres $\pm\mathbf{e}_j$ to face centres, so each column of Q is some $\pm\mathbf{e}_i$: Q is a *signed permutation matrix*, a permutation of the three axes combined with a choice of sign for each. There are $3! \times 2^3 = 48$ of them, the rotations of the

cube ($\det Q = +1$) and their products with the inversion $\boldsymbol{\xi} \mapsto -\boldsymbol{\xi}$ ($\det Q = -1$), which include the reflections in the coordinate and diagonal planes. The product of two symmetries and the inverse $Q^{-1} = Q^{\mathsf{T}}$ of one are again symmetries, so they form a group, the octahedral group O_h. For example, the quarter turn about axis 0 and the reflection in the plane $\xi_0 = 0$ are

$$Q_{\text{turn}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad Q_{\text{refl}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{25}$$

How a symmetry acts on the field of the cell. Applying Q to a field carries the value at $\boldsymbol{\xi}$ to $Q\boldsymbol{\xi}$ and turns its components with it: the displacement is a vector, $u \mapsto Qu$, and the strain a symmetric tensor, $\varepsilon \mapsto Q\varepsilon Q^{\mathsf{T}}$, that is $\varepsilon'_{ij} = \sum_{k,l} Q_{ik} Q_{jl} \varepsilon_{kl}$. On the nine components $\psi = (u, \varepsilon)$ this is the 9×9 matrix

$$S(Q) = \begin{pmatrix} Q & 0 \\ 0 & \Sigma(Q) \end{pmatrix}, \tag{26}$$

with $\Sigma(Q)$ the 6×6 matrix of $\varepsilon \mapsto Q\varepsilon Q^{\mathsf{T}}$ in Voigt form. The transformed field is $\psi'(\boldsymbol{\xi}) = S(Q)\,\psi(Q^{\mathsf{T}}\boldsymbol{\xi})$. For the cell's field $\psi(\boldsymbol{\xi}) = \sum_b \psi_b L_b(\boldsymbol{\xi})$ the functions are moved as well: 1 is unchanged, and $L_k(Q^{\mathsf{T}}\boldsymbol{\xi}) = (Q^{\mathsf{T}}\boldsymbol{\xi})_k = \sum_j Q_{jk}\,\xi_j$ for the linear functions $L_k = \xi_k$. So the four functions are mixed by the 4×4 matrix

$$T(Q) = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}, \tag{27}$$

and the coefficients of the transformed field are

$$\psi'_a = \sum_{b=0}^{3} T_{ab}(Q)\, S(Q)\, \psi_b, \qquad a = 0, \dots, 3. \tag{28}$$

Two things happen at once and independently: $T(Q)$ mixes *which function* carries the field, and $S(Q)$ mixes *which component* of the field it is.

The Kronecker product. Stack the 36 unknowns as one column, $\Psi = (\psi_0, \psi_1, \psi_2, \psi_3)$, each ψ_b nine components, so that the unknown with function index a and component i sits in row $9a + i$. Eq. (28) is then $\Psi' = D(Q)\Psi$ with the 36×36 matrix

$$D(Q) = T(Q) \otimes S(Q) = \begin{pmatrix} T_{00}S & T_{01}S & T_{02}S & T_{03}S \\ T_{10}S & T_{11}S & T_{12}S & T_{13}S \\ T_{20}S & T_{21}S & T_{22}S & T_{23}S \\ T_{30}S & T_{31}S & T_{32}S & T_{33}S \end{pmatrix}, \qquad D(Q)_{9a+i,\,9b+j} = T_{ab}(Q)\, S_{ij}(Q), \tag{29}$$

the Kronecker product: the 4×4 array of 9×9 blocks whose block (a, b) is $S(Q)$ scaled by the number $T_{ab}(Q)$. It is the matrix of a map that acts on the function index by T and on the component index by S, each without regard to the other. For the reflection of Eq. (25), $T = \mathrm{diag}(1, -1, 1, 1)$, and S is diagonal with -1 on u_0 and on the shears ε_{01} and ε_{02}, which change sign with ξ_0, and $+1$ elsewhere; so D is diagonal, and the sign of the unknown $\xi_0 u_0$, for instance,

is $T_{11}S_{00} = (-1)(-1) = +1$. Because the Kronecker product satisfies $(A \otimes B)(C \otimes F) = AC \otimes BF$, and T and S each respect products,

$$D(Q_1)\,D(Q_2) = D(Q_1Q_2) \qquad \text{for all } Q_1, Q_2 \in O_h : \tag{30}$$

the assignment $Q \mapsto D(Q)$ turns composition of symmetries into multiplication of matrices, which is what is meant by a *representation* of the group. For a signed permutation, $Q\varepsilon Q^{\mathsf{T}}$ permutes the strain components with signs and sends shears to shears, so $\Sigma(Q)$, $S(Q)$, $T(Q)$ and $D(Q)$ are all signed permutation matrices, and orthogonal: $D(Q)^{-1} = D(Q)^{\mathsf{T}} = D(Q^{\mathsf{T}})$.

Why the cell's matrices commute with it. Three facts make

$$D(Q)\,\mathbf{K}\,D(Q)^{\mathsf{T}} = \mathbf{K}, \qquad D(Q)\,\mathbf{M}\,D(Q)^{\mathsf{T}} = \mathbf{M}, \qquad D(Q)\,\mathbf{\Delta}\,D(Q)^{\mathsf{T}} = \mathbf{\Delta} \qquad \text{for all 48 } Q. \tag{31}$$

The domain is invariant: Q maps the cube onto itself. *The kernel is covariant*: the background is isotropic, so its Green's tensor satisfies $\mathbf{G}(Q\mathbf{r}) = Q\,\mathbf{G}(\mathbf{r})\,Q^{\mathsf{T}}$ for every orthogonal Q, reflections included, and its derivatives transform as tensors of the corresponding rank; changing variables $\mathbf{x} \mapsto Q\mathbf{x}$, $\mathbf{y} \mapsto Q\mathbf{y}$ in the double integral of $\mathbf{K}$ then gives the first identity. *The contrast is invariant*: uniform in the cell, $\omega^2\Delta\rho$ is a scalar and δc an isotropic tensor, so $\mathbf{\Delta}$ commutes with $S(Q)$; and the Gram matrix $\mathbf{M}$ only weights the functions 1 and ξ_i, which Q permutes among themselves with equal weights. The system matrix $\mathbf{M} - \mathbf{K}\mathbf{\Delta}$, its inverse, and $\mathbf{T}_{36}$ of Eq. (7) therefore commute with every $D(Q)$. Nothing here requires $\mathbf{K}$ to be Hermitian or real: the identities hold for the complex self block at every frequency, radiation included, and power by power for each $\mathbf{K}_n$.

Schur's lemma. A representation is *irreducible* when no subspace other than 0 and the whole is mapped into itself by every $D(Q)$. Schur's lemma states that a linear map between two irreducible representations that commutes with the group action is zero if the representations are inequivalent, and a multiple of the identity if they are the same. The 36-dimensional representation of Eq. (29) is not irreducible; it is a direct sum of copies of the ten irreducible representations of O_h, labelled by their dimension (A: 1, E: 2, T: 3) and by their parity under the inversion $\boldsymbol{\xi} \mapsto -\boldsymbol{\xi}$ (g even, u odd). How many copies n_Γ of each Γ occur follows from the characters, the traces of the representation matrices: $n_\Gamma = \frac{1}{48}\sum_Q \chi_\Gamma(Q)\,\mathrm{tr}\,D(Q)$, with $\mathrm{tr}\,D(Q) = \mathrm{tr}\,T(Q)\,\mathrm{tr}\,S(Q)$. Every irreducible representation of O_h can be written with real matrices and has real characters, so the lemma holds over the real numbers, and applies to the real and imaginary parts of $\mathbf{K}$ alike. Applied to an operator that commutes with $D(Q)$, it has two consequences. First, the operator does not couple unknowns of different Γ. Second, within the n_Γ copies of a Γ of dimension d_Γ, if a basis is chosen in which every copy is represented by the *same* matrices (each basis vector a *partner*, transforming as a given row of Γ), the operator acts as an $n_\Gamma \times n_\Gamma$ matrix on the multiplicity index and as the identity on the row index, which is Eq. (14), and likewise for $\mathbf{M}$, $\mathbf{\Delta}$, $\mathbf{M} - \mathbf{K}\mathbf{\Delta}$ and $\mathbf{T}_{36}$. The block $\mathbf{K}_\Gamma$ is then found from any one row: with V_Γ the $36 \times n_\Gamma$ matrix whose columns are the partners of that row, $\mathbf{K}V_\Gamma = V_\Gamma\mathbf{K}_\Gamma$. Inverting the 36×36 system reduces to inverting $n_\Gamma \times n_\Gamma$ blocks, of size at most 4×4.

The partners. Table 6 gives the partners of one row of each Γ as integer vectors. They are chosen so that within each Γ they transform alike: the displacement's first moment $\xi_1 u_2 + \xi_2 u_1$ and the shear $2\varepsilon_{12}$, for instance, both transform as the $(1,2)$ row of T_{2g}. The reduction is then exact: $\mathbf{K}_n V_\Gamma = V_\Gamma \mathbf{K}_{\Gamma,n}$ with no residual at any power computed (§6).

Table 6. One row of each irreducible representation, as combinations of the unknowns $\xi_i u_j$ ($L_b = \xi_i$ times the displacement component j), u_j, ε_{ij} and $\xi_i \varepsilon_{jk}$, axes $0, 1, 2$.

Γ	vectors	M_Γ/h^3
A_{1g}	$\xi_0 u_0 + \xi_1 u_1 + \xi_2 u_2$; $\varepsilon_{00} + \varepsilon_{11} + \varepsilon_{22}$	$\frac{8}{3}, 8$
E_g	$2\xi_0 u_0 - \xi_1 u_1 - \xi_2 u_2$; $2\varepsilon_{00} - \varepsilon_{11} - \varepsilon_{22}$	$\frac{8}{3}, 8$
T_{1g}	$\xi_1 u_2 - \xi_2 u_1$	$\frac{8}{3}$
T_{2g}	$\xi_1 u_2 + \xi_2 u_1$; $2\varepsilon_{12}$	$\frac{8}{3}, 8$
T_{1u}	u_0; $\xi_0\varepsilon_{00}$; $\xi_0(\varepsilon_{11} + \varepsilon_{22})$; $2(\xi_1\varepsilon_{01} + \xi_2\varepsilon_{02})$	$8, \frac{8}{3}, \frac{8}{3}, \frac{8}{3}$
T_{2u}	$\xi_0(\varepsilon_{11} - \varepsilon_{22})$; $2(\xi_1\varepsilon_{01} - \xi_2\varepsilon_{02})$	$\frac{8}{3}, \frac{8}{3}$
A_{2u}	$2(\xi_0\varepsilon_{12} + \xi_1\varepsilon_{02} + \xi_2\varepsilon_{01})$	$\frac{8}{3}$
E_u	$2(\xi_1\varepsilon_{02} - \xi_2\varepsilon_{01})$	$\frac{8}{3}$

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